

Day 4: Hands-on Subnetting Practical :Divide a network into subnets and assign IP addresses to devices. Task: Document the steps to create a subnet plan.

Tasks :-

1. Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Ans. :-

Step 1: Write the Given Network

- **Network Address:** 192.168.10.0
- **CIDR:** /24
- **Subnet Mask:** 255.255.255.0

A /24 network has 256 IP addresses (from 192.168.10.0 to 192.168.10.255).

Step 2: Find How Many Bits to Borrow :-

We need 4 equal subnets.

Use the formula:

$2^n = \text{Number of Subnets}$

Where **n = borrowed bits**

- $2^2 = 4$

So, borrow 2 bits.

Step 3: Find the New Subnet Mask :-

A /26 prefix gives the subnet mask:

255.255.255.192

Step 4: Calculate the Block Size :-

Use this formula:

Block Size = 256 – Last Octet of the Subnet Mask

= 256 – 192

= **64**

Subnet	Subnet Mask	Network Address	Broadcast Address
1	255.255.255.192 (/26)	192.168.10.0	192.168.10.63
2	255.255.255.192 (/26)	192.168.10.64	192.168.10.127
3	255.255.255.192 (/26)	192.168.10.128	192.168.10.191
4	255.255.255.192 (/26)	192.168.10.192	192.168.10.255

**2. For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Hint: Make sure not to use the network or broadcast address for any device.**

Ans.:-

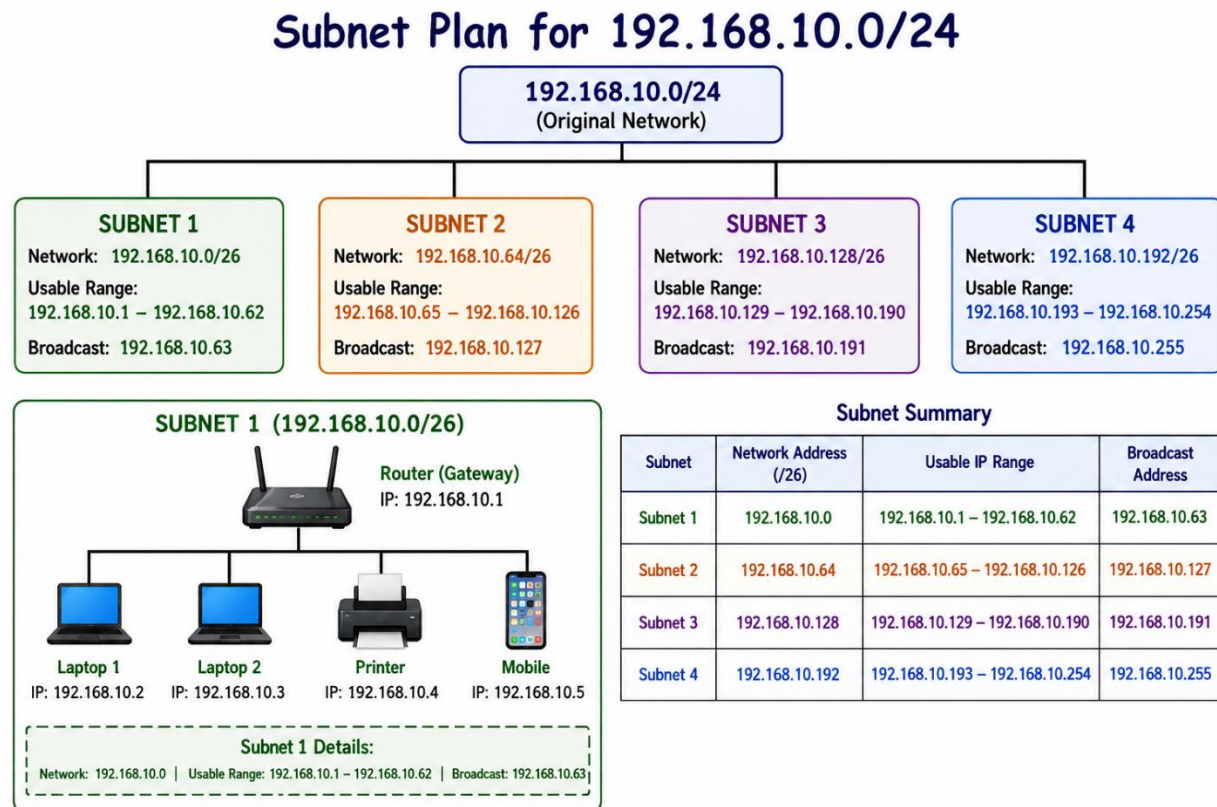
Using Subnet 1:-

- Network Address: 192.168.10.0/26
- Subnet Mask: 255.255.255.192
- Usable Host Range: 192.168.10.1 – 192.168.10.62
- Broadcast Address: 192.168.10.63

Device	Assigned IP Address
Router	192.168.10.1
Laptop 1	192.168.10.2
Laptop 2	192.168.10.3
Printer	192.168.10.4
Mobile	192.168.10.5

3. Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Ans.:-



4. Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Ans. :-

Step 1: Understand the Given Network:-

The given network is **192.168.10.0/24**.

- /24 means the first 24 bits are used for the network.
- The subnet mask is 255.255.255.0.
- A /24 network contains 256 IP addresses (0–255).

Step 2: Divide the Network into 4 Equal Subnets:-

We need 4 subnets.

To create 4 subnets, we borrow **2 bits** from the host portion because:

- $2^2 = 4$ subnets

The new prefix becomes:

- $/24 + 2 = /26$

New subnet mask:

- **255.255.255.192**

Step 3: Calculate the Number of Hosts:-

A /26 subnet leaves 6 host bits.

Number of IP addresses:

- $2^6 = 64$ IP addresses

Usable host addresses:

- $64 - 2 = 62$ hosts

Subnet	Network Address	Usable Host Range	Broadcast Address
1	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129 – 192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193 – 192.168.10.254	192.168.10.255